

CS202: Software Tools and Techniques for CSE

Lecture 3

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Source code *diff* (unified)

github.com/SET-IITGN/cs202/commit/45a77e06f0b1b13907fcca80e7debd614fe0ba3f?diff=unified#diff-5cc5f480c15364f8b27b8941ae1e9498d3170d3a6e13421e7d527e7e1b5e5410

Filter files...

- src.c
- vuln.c

2 files changed +1 -2 lines changed

Search within code

src.c +1 -1

```
@@ -12,7 +12,7 @@ int main() {
12 12     char *buffer;
13 13     char message[] = "Hello, worlcome to CS202";
14 14
15 -     int numBytes = strlen(message);
15 +     int numBytes = strlen(message);
16 16     buffer=(char*)malloc(numBytes+1);
17 17     memset(buffer, '\0', numBytes+1);
18 18     copyMemory(buffer, message, numBytes);
****
↓
```

vuln.c -1

```
... @@ -1,4 +1,3 @@
1 - //base file
2 1     #include <stdio.h>
3 2
4 3     void copyMemory(char* destination, const char* source, int numBytes) {
****
↓
```

Source code *diff* (split)

github.com/SET-IITGN/cs202/commit/45a77e06f0b1b13907fcca80e7debd614fe0ba3f?diff=split#diff-5cc5f480c15364f8b27b8941ae1e9498d3170d3a6e13421e7d527e7e1b5e5410

Filter files...

src.c
vuln.c

2 files changed +1 -2 lines changed

Search within code

src.c +1 -1 ...

```
@@ -12,7 +12,7 @@ int main() {
12     char *buffer;
13     char message[] = "Hello, worlcome to CS202";
14
15 -     int numBytes = strlen(message);
16     buffer=(char*)malloc(numBytes+1);
17     memset(buffer, '\0', numBytes+1);
18     copyMemory(buffer, message, numBytes);
```

vuln.c -1 ...

```
@@ -1,4 +1,3 @@
1 - //base file
2     #include <stdio.h>
3
4     void copyMemory(char* destination, const char*
        source, int numBytes) {
```

Browsing a commit on GitHub

Navigation bar: <> Code Issues 1 Pull requests Actions Projects Wiki Security 3 Insights ...

Commit 45a77e0 (highlighted in red box)

shouvick149 committed 2 weeks ago · 4 / 5

A few more minor edits.

main (highlighted in blue box)

1 parent 505adce commit 45a77e0 (highlighted in red box)

old version (v1) (highlighted in blue box) → 505adce

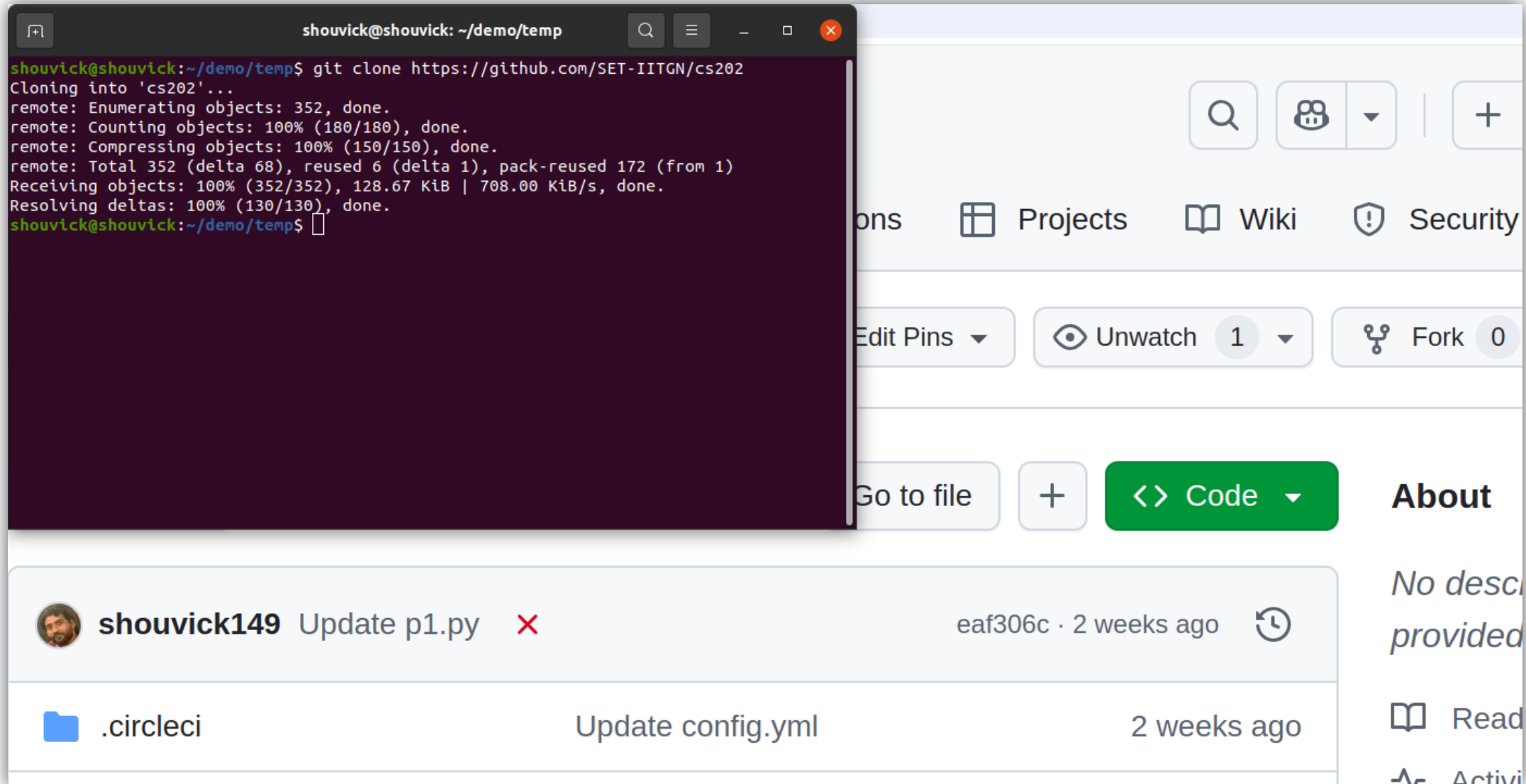
new version (v2) (highlighted in red box) → 45a77e0

2 files changed: +1 -2 lines changed

src.c, vuln.c

```
@@ -12,7 +12,7 @@ int main() {
12 char *buffer;
```

git clone https://github.com/SET-IITGN/cs202



The image shows a terminal window in the foreground and a GitHub repository page in the background. The terminal window, titled 'shouvick@shouvick: ~/demo/temp', displays the output of the command 'git clone https://github.com/SET-IITGN/cs202'. The output shows the cloning process, including enumerating, counting, and compressing objects, and receiving and resolving deltas. The terminal window is partially obscuring the GitHub page, which shows the repository 'shouvick149' with a commit 'eaf306c' from 2 weeks ago. The repository page also shows a file '.circleci' with a commit 'Update config.yml' from 2 weeks ago. The GitHub page includes navigation links like 'Projects', 'Wiki', and 'Security', and a 'Code' button.

```
shouvick@shouvick:~/demo/temp$ git clone https://github.com/SET-IITGN/cs202
Cloning into 'cs202'...
remote: Enumerating objects: 352, done.
remote: Counting objects: 100% (180/180), done.
remote: Compressing objects: 100% (150/150), done.
remote: Total 352 (delta 68), reused 6 (delta 1), pack-reused 172 (from 1)
Receiving objects: 100% (352/352), 128.67 KiB | 708.00 KiB/s, done.
Resolving deltas: 100% (130/130), done.
shouvick@shouvick:~/demo/temp$
```

shouvick149 Update p1.py ✖ eaf306c · 2 weeks ago 

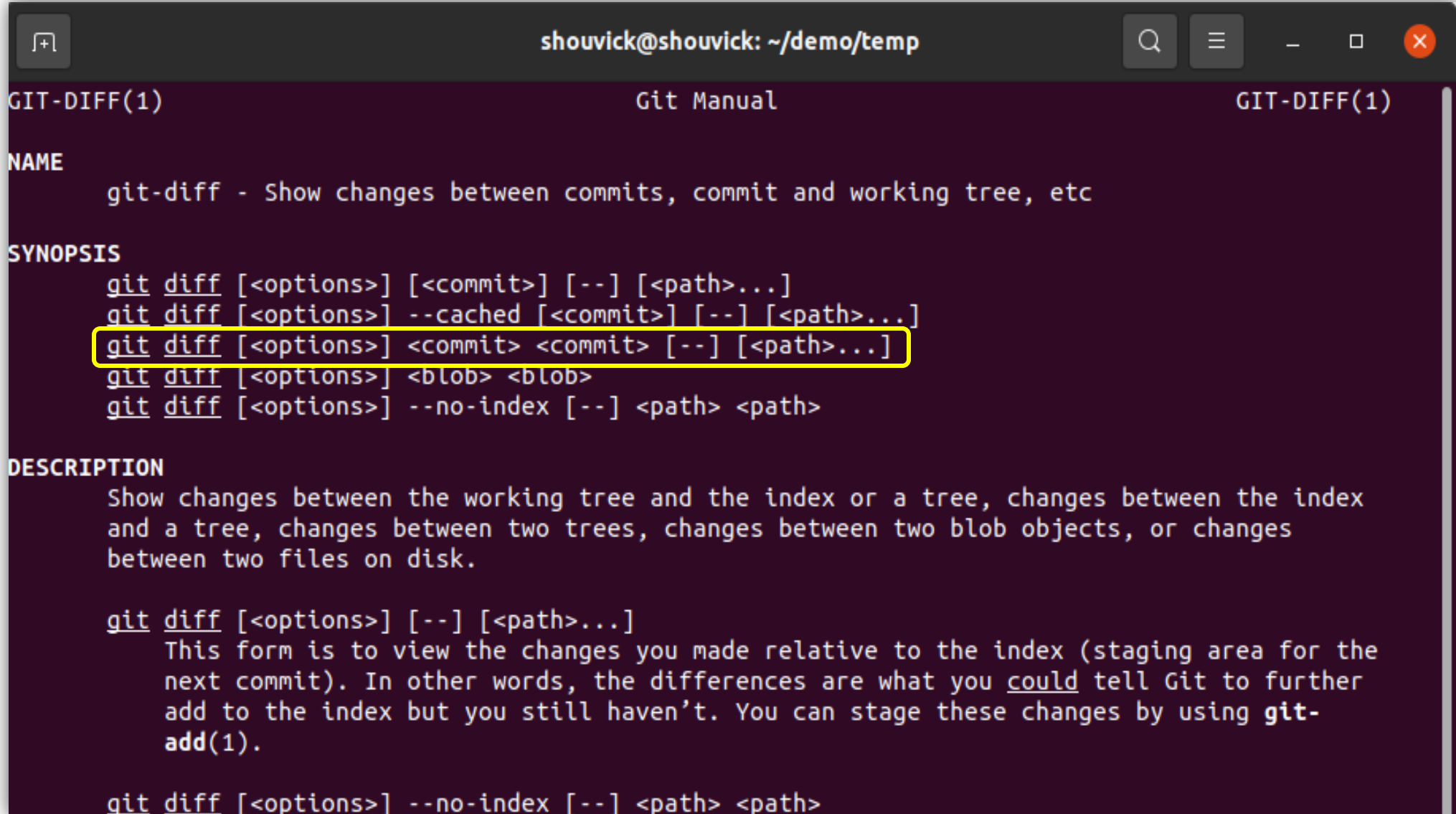
 .circleci Update config.yml 2 weeks ago

Go to file + <> Code About

No description provided

Readme  Activity

cd cs202; git diff --help

A terminal window with a dark background and light text. The title bar shows the user 'shouvick@shouvick' and the directory '~/demo/temp'. The terminal output is the help text for 'git diff'. The third line of the synopsis, 'git diff [<options>] <commit> <commit> [--] [<path>...]', is highlighted with a yellow box. The window has standard Linux window controls (minimize, maximize, close) in the top right corner.

```
GIT-DIFF(1)                               Git Manual                               GIT-DIFF(1)

NAME
    git-diff - Show changes between commits, commit and working tree, etc

SYNOPSIS
    git diff [<options>] [<commit>] [--] [<path>...]
    git diff [<options>] --cached [<commit>] [--] [<path>...]
    git diff [<options>] <commit> <commit> [--] [<path>...]
    git diff [<options>] <blob> <blob>
    git diff [<options>] --no-index [--] <path> <path>

DESCRIPTION
    Show changes between the working tree and the index or a tree, changes between the index
    and a tree, changes between two trees, changes between two blob objects, or changes
    between two files on disk.

    git diff [<options>] [--] [<path>...]
    This form is to view the changes you made relative to the index (staging area for the
    next commit). In other words, the differences are what you could tell Git to further
    add to the index but you still haven't. You can stage these changes by using git-
    add(1).

    git diff [<options>] --no-index [--] <path> <path>
```

Source code *diff* (unified)

github.com/SET-IITGN/cs202/commit/45a77e06f0b1b13907fcca80e7debd614fe0ba3f?diff=unified#diff-5cc5f480c15364f8b27b8941ae1e9498d3170d3a6e13421e7d527e7e1b5e5410

Filter files... src.c vuln.c

2 files changed +1 -2 lines changed

Search within code

src.c

```
@@ -12,7 +12,7 @@ int main() {
12 12 char *buffer;
13 13 char message[] = "Hello, worlcome to CS202";
14 14
15 - int numBytes = strlen(message);
15 + int numBytes = strlen(message);
16 16 buffer=(char*)malloc(numBytes+1);
17 17 memset(buffer, '\0', numBytes+1);
18 18 copyMemory(buffer, message, numBytes);
```

vuln.c

```
@@ -1,4 +1,3 @@
1 - //base file
2 1 #include <stdio.h>
3 2
4 3 void copyMemory(char* destination, const char* source, int numBytes) {
```

git diff <parent_commit> <curr_commit>

```
diff --git a/src.c b/src.c
index 0d6a029..85bb4fb 100644
--- a/src.c
+++ b/src.c
@@ -12,7 +12,7 @@ int main() {
     char *buffer;
     char message[] = "Hello, worlcome to CS202";

-    int numBytes = strlen(message);
+    int numBytes = strlen(message);
     buffer=(char*)malloc(numBytes+1);
     memset(buffer, '\0', numBytes+1);
     copyMemory(buffer, message, numBytes);
diff --git a/vuln.c b/vuln.c
index 838cba5..c1a0e33 100644
--- a/vuln.c
+++ b/vuln.c
@@ -1,4 +1,3 @@
-//base file
#include <stdio.h>
```


Empirical Software Engineering (2020) 25:790–823

<https://doi.org/10.1007/s10664-019-09772-z>

How different are different *diff* algorithms in Git?

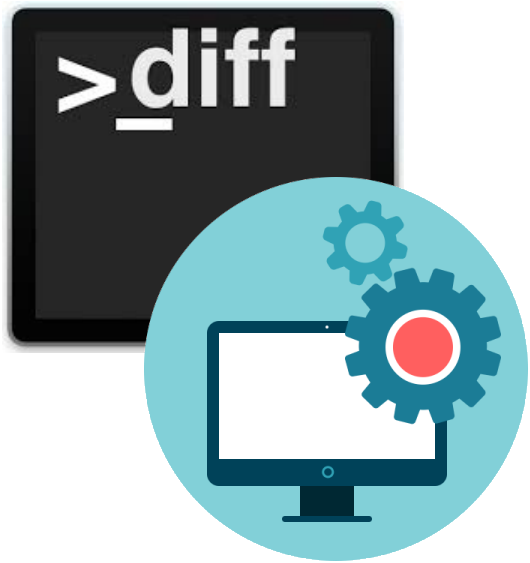
Use `--histogram` for code changes

Yusuf Sulistyo Nugroho¹  · Hideaki Hata¹ · Kenichi Matsumoto¹

Published online: 11 September 2019

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diff is essential in SE research field



**Empirical Software
Engineering Research**



GitHub

```
git diff [<options>] <commit> <commit> [--] [<path>...]
```

Git offers 4 diff algorithms



Documentation

`--diff-algorithm={algorithm name}`

- **Myers** default algorithm
- **Minimal** (improved Myers)
- **Patience** (try to give contextual diff)
- **Histogram** (enhanced Patience, normally faster)

Histogram was introduced in git 1.7.7 in 2011

Tree: 77bd3ea9f5 [git / Documentation / RelNotes / 1.7.7.txt](#)

Latest commit 783f05a on Oct 1, 2011 [History](#)

134 lines (95 sloc) | 5.28 KB [Raw](#) [Blame](#) [Icons](#)

```
1  Git v1.7.7 Release Notes
2  =====
3
4  Updates since v1.7.6
5  -----
6
7  * The scripting part of the codebase is getting prepared for i18n/l10n.
8
9  * Interix, Cygwin and Minix ports got updated.
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67
68  * "git diff" learned a "--histogram" option to use a different diff
69  generation machinery stolen from jgit, which might give better
70  performance.
71
72  * "git diff" had a weird worst case behaviour that can be triggered
73  when comparing files with potentially many places that could match.
74
```

Different algorithms produce different *diff* outputs

```
1 @@ -169,11 +284,36 @@ public class AmqpMessage {
2     * @throws IllegalStateException if the message is read only.
3
4     9 added lines
5     + checkReadOnly();
6     + AmqpValue body = new AmqpValue(value);
7     + getWrappedMessage().setBody(body);
8     + }
9     +
10    + //----- Internal implementation -----
11    +
12    + private void checkReadOnly() throws IllegalStateException {
13        if (delivery != null) {
14            throw new IllegalStateException("Message is read only.");
15        }
16    + }
17
18    - AmqpValue body = new AmqpValue(value);
19    - getWrappedMessage().setBody(body);
```

2 deleted lines

a) **diff** extraction using Myers

Differences:

- Number of changed lines
- Position of changed lines

```
1 @@ -169,11 +284,36 @@ public class AmqpMessage {
2     * @throws IllegalStateException if the message is read only.
3
4     4 deleted lines
5     - if (delivery != null) {
6     -     throw new IllegalStateException("Message is read only.");
7     - }
8     -
9     + checkReadOnly();
10    AmqpValue body = new AmqpValue(value);
11    getWrappedMessage().setBody(body);
12    }
13    +
14    + //----- Internal implementation -----
15    +
16    + private void checkReadOnly() throws IllegalStateException {
17    +     if (delivery != null) {
18    +         throw new IllegalStateException("Message is read only.");
19    +     }
20    + }
```

9 added lines

b) **diff** extraction using Histogram

https://github.com/ome/openmicroscopy

The screenshot shows the GitHub interface for a specific commit. At the top, the URL is `https://github.com/ome/openmicroscopy/commit/844e0fde447d2d069fb17c480e95acf4d372afc4#diff-07322c93ef4fb3f0dd245932b74b10e1`. The repository name is `ome / openmicroscopy`. The commit is titled **Commit 844e0fd** and was made by `bwzloranger` on August 4, 2010. The commit message is `Fixes for importer findbugs`, with a `git-svn-id` provided below it. The commit is linked to the `develop` branch and the `v5.6.14` tag. A file tree on the left shows the directory structure, with `components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java` highlighted. The main area shows a diff for 7 files, with `GuiCommonElements.java` selected. The diff shows a change in the `addComboBox` method, with a red box highlighting the new line of code.

ome / openmicroscopy

Search Type / to search

<> Code Issues 26 Pull requests 6 Actions Projects Wiki Security Insights

Commit 844e0fd

Preview Give feedback Browse files

bwzloranger committed on Aug 4, 2010

Fixes for importer findbugs

git-svn-id: file:///home/svn/omero/trunk@7668 05709c45-44f0-0310-885b-81a1db45b4a6

develop · v5.6.14 ... 4_2_START 1 parent cb85815 commit 844e0fd

Filter files...

- components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
- FileQueueHandler.java
- FileQueueTable.java
- GuiCommonElements.java
- HistoryTable.java
- HistoryTableAbstractDataS...

7 files changed +114 -88 lines changed

Search within code

- components/tools/OmeroImporter/src/ome/formats/importer/gui/FileQueueHandler.java +18 -18
- components/tools/OmeroImporter/src/ome/formats/importer/gui/FileQueueTable.java +77 -49
- components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java +11 -7

```
@@ -673,17 +673,21 @@ public static JComboBox addComboBox(Container container, String label,
```

git diff --diff-algorithm=myers ...

```
components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
@@ -673,17 +673,21 @@ public static JComboBox addComboBox(Container container, S
673 673      */
674 674      public static ImageIcon getImageIcon(String path)
675 675      {
676 +          if (path == null)
677 +          {
678 +              log.error("Icon path is null");
679 +              return null;
680 +          }
681 +
682      java.net.URL imgURL = GuiImporter.class.getResource(path);
683
684 -          if (imgURL != null)
685 +          if (imgURL == null)
686 +          {
687 -              return new ImageIcon(imgURL);
688 -          }
689 -          else
690 -          {
691 -              log.error("Couldn't find icon: " + imgURL);
692 +              log.error("Couldn't find icon: " + imgURL);
693 +              return null;
694 +          }
695 +          return new ImageIcon(imgURL);
696      }
697  }
```

```
shouvik@shouvik:~/demo/openmicroscopy$ git diff --diff-algorithm=my
nts/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElement
diff --git a/components/tools/OmeroImporter/src/ome/formats/importer/
omponents/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonE
index 244701eb0e..d783fb51b6 100644
--- a/components/tools/OmeroImporter/src/ome/formats/importer/gui/Gui
+++ b/components/tools/OmeroImporter/src/ome/formats/importer/gui/Gui
@@ -673,17 +673,21 @@ public class GuiCommonElements
    */
    public static ImageIcon getImageIcon(String path)
    {
+        if (path == null)
+        {
+            log.error("Icon path is null");
+            return null;
+        }
        java.net.URL imgURL = GuiImporter.class.getResource(path);

-        if (imgURL != null)
+        if (imgURL == null)
        {
            return new ImageIcon(imgURL);
        }
        else
        {
            log.error("Couldn't find icon: " + imgURL);
+            log.error("Couldn't find icon: " + imgURL);
            return null;
        }
        else
        {
            return new ImageIcon(imgURL);
        }
    }
    /**
```

git diff -w --ignore-blank-lines --diff-algorithm=myers

Version 1

```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
676 -     java.net.URL imgURL = GuiImporter.class.getResource(path);
677 -
678 -     if (imgURL != null)
679     {
680 -         return new ImageIcon(imgURL);
681     }
682 - else
683     {
684         log.error("Couldn't find icon: " + imgURL);
685 -     }
686     return null;
687 }
688 /**
689 ... other code here ...
```

Version 2

```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
676 +     if (path == null)
677     {
678 +         log.error("Icon path is null");
679 +         return null;
680     }
681 +
682 +     java.net.URL imgURL = GuiImporter.class.getResource(path);
683 +
684 +     if (imgURL == null)
685     {
686         log.error("Couldn't find icon: " + imgURL);
687     }
688     return null;
689 + else
690 +     return new ImageIcon(imgURL);
691 + }
692 /**
693 ... other code here ...
```

After pairing up all matching lines...

git diff -w --ignore-blank-lines --diff-algorithm=myers

```
shouvick@shouvick:~/demo/openmicroscopy$ git diff -w --ignore-blank-lines --diff-algorithm=myers cb85815 84
Elements.java
diff --git a/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java b/component
index 244701eb0e..d783fb51b6 100644
--- a/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
+++ b/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
@@ -673,18 +673,22 @@ public class GuiCommonElements
    */
    public static ImageIcon getImageIcon(String path)
    {
-        java.net.URL imgURL = GuiImporter.class.getResource(path);
-
-        if (imgURL != null)
+        if (path == null)
+        {
-            return new ImageIcon(imgURL);
+            return new ImageIcon(imgURL);
+            log.error("Icon path is null");
+            return null;
+        }
-        else
+        else
+        {
+            java.net.URL imgURL = GuiImporter.class.getResource(path);
+            if (imgURL == null)
+            {
+                log.error("Couldn't find icon: " + imgURL);
+            }
+            return null;
+        }
+        else
+        {
+            return new ImageIcon(imgURL);
+        }
    }

    /**
     * Quit Confirmation pop up dialog

```


... -w --ignore-blank-lines --diff-algorithm=histogram

```
shouvick@shouvick:~/demo/openmicroscopy$ git diff -w --ignore-blank-lines --diff-algorithm=histogram cb85815
commonElements.java
diff --git a/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java b/components
index 244701eb0e..d783fb51b6 100644
--- a/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
+++ b/components/tools/OmeroImporter/src/ome/formats/importer/gui/GuiCommonElements.java
@@ -673,18 +673,22 @@ public class GuiCommonElements
    */
    public static ImageIcon getImageIcon(String path)
    {
+   if (path == null)
+   {
+       log.error("Icon path is null");
+       return null;
+   }
    java.net.URL imgURL = GuiImporter.class.getResource(path);

    if (imgURL != null)
    {
        return new ImageIcon(imgURL);
    }
    else
+   if (imgURL == null)
    {
        log.error("Couldn't find icon: " + imgURL);
    }
    return null;
+   }
+   else
+   return new ImageIcon(imgURL);
+   }

    /**
     * Quit Confirmation pop up dialog
```

... -w --ignore-blank-lines --diff-algorithm=histogram

Version 1

```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
    java.net.URL imgURL = GuiImporter.class.getResource(path);
    if (imgURL != null)
    {
        return new ImageIcon(imgURL);
    }
    else
    {
        log.error("Couldn't find icon: " + imgURL);
    }
    return null;
}

/**
... other code here ...
```

Version 2

```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
676 +     if (path == null)
677 +     {
678 +         log.error("Icon path is null");
679 +         return null;
680 +     }
681 +
682     java.net.URL imgURL = GuiImporter.class.getResource(path);
683
684 +     if (imgURL == null)
685     {
686         log.error("Couldn't find icon: " + imgURL);
687         return null;
688     }
689 +     else
690 +         return new ImageIcon(imgURL);
691 + }

/**
... other code here ...
```

After pairing up all matching unique lines... (think of alignment)

... -w --ignore-blank-lines --diff-algorithm=myers

Version 1

```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
676 - java.net.URL imgURL = GuiImporter.class.getResource(path);
677 -
678 - if (imgURL != null)
679 {
680 - return new ImageIcon(imgURL);
681 }
682 - else
683 {
684     log.error("Couldn't find icon: " + imgURL);
685 - }
686     return null;
687 }
688
689 /**
... other code here ...
```

Version 2

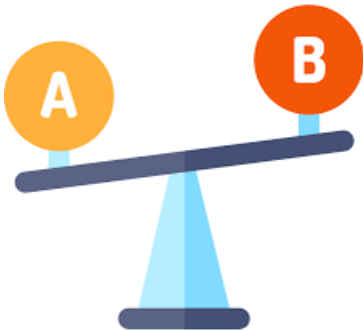
```
... other code here ...
673 */
674 public static ImageIcon getImageIcon(String path)
675 {
676 + if (path == null)
677 {
678 +     log.error("Icon path is null");
679 +     return null;
680 }
681 +
682 + java.net.URL imgURL = GuiImporter.class.getResource(path);
683 +
684 + if (imgURL == null)
685 {
686     log.error("Couldn't find icon: " + imgURL);
687     return null;
688 }
689 + else
690 +     return new ImageIcon(imgURL);
691 + }
692
693 /**
... other code here ...
```

After pairing up all matching lines...

Two sequential analyses

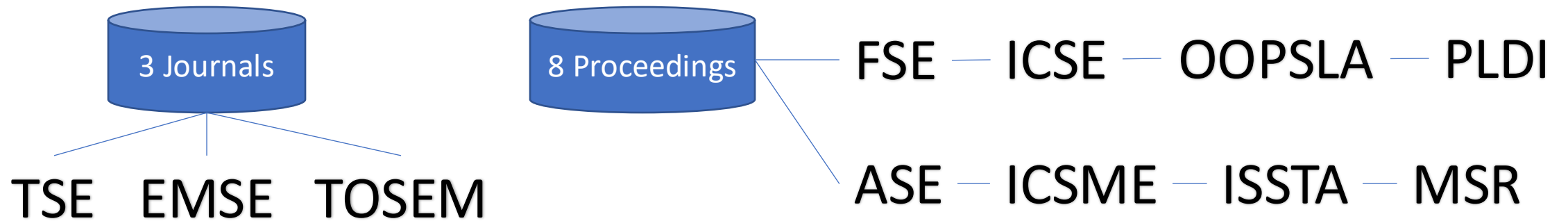


- Systematic Mapping Study
 - How previous studies used git diff?

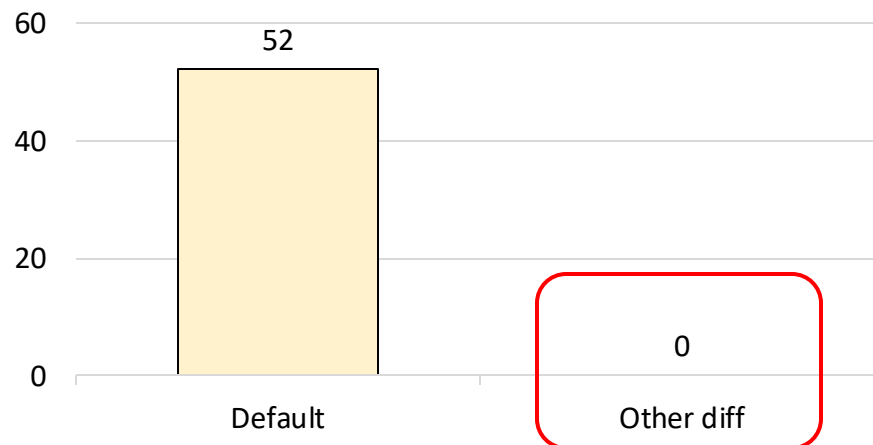


- Comparisons Study
 - differences of *diff* outputs between Myers and Histogram

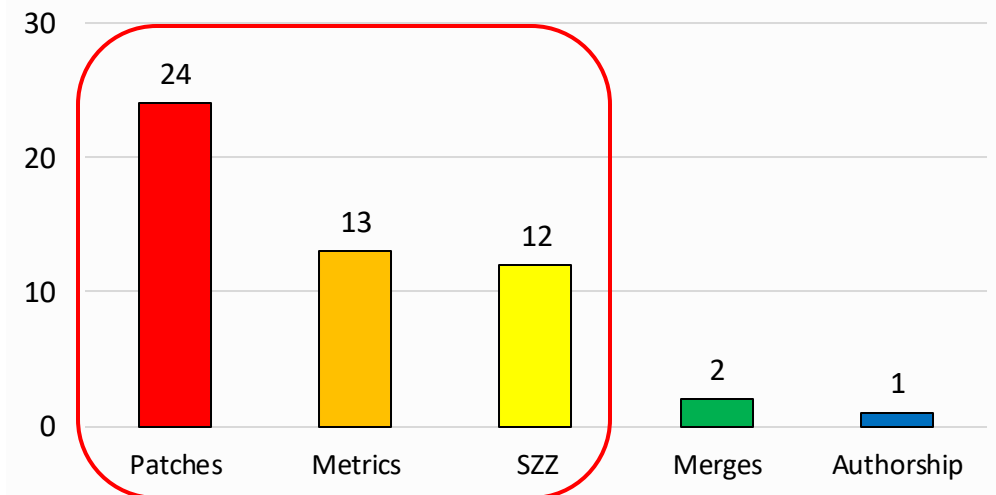
Results of Mapping Study



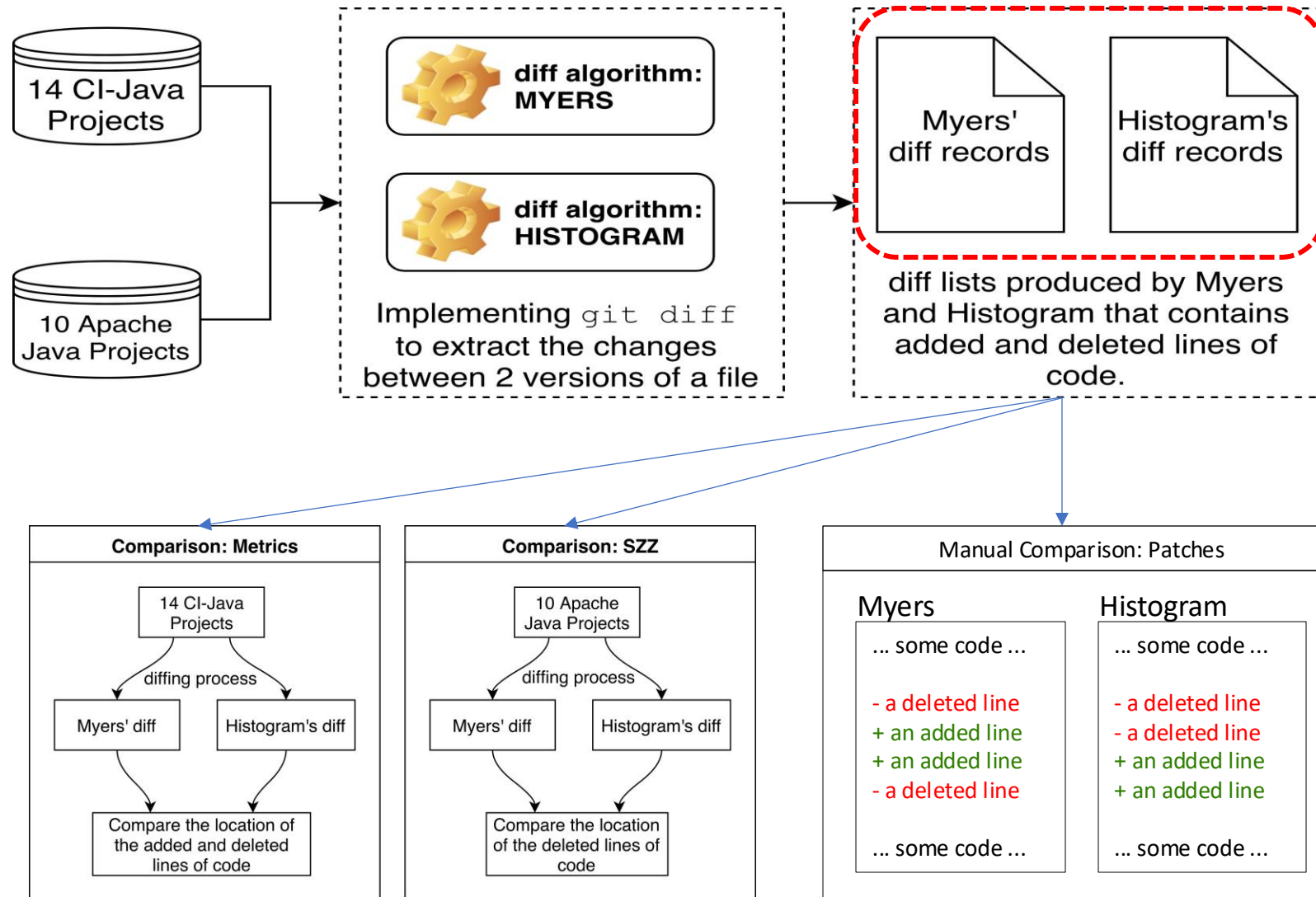
Type of diff algorithm

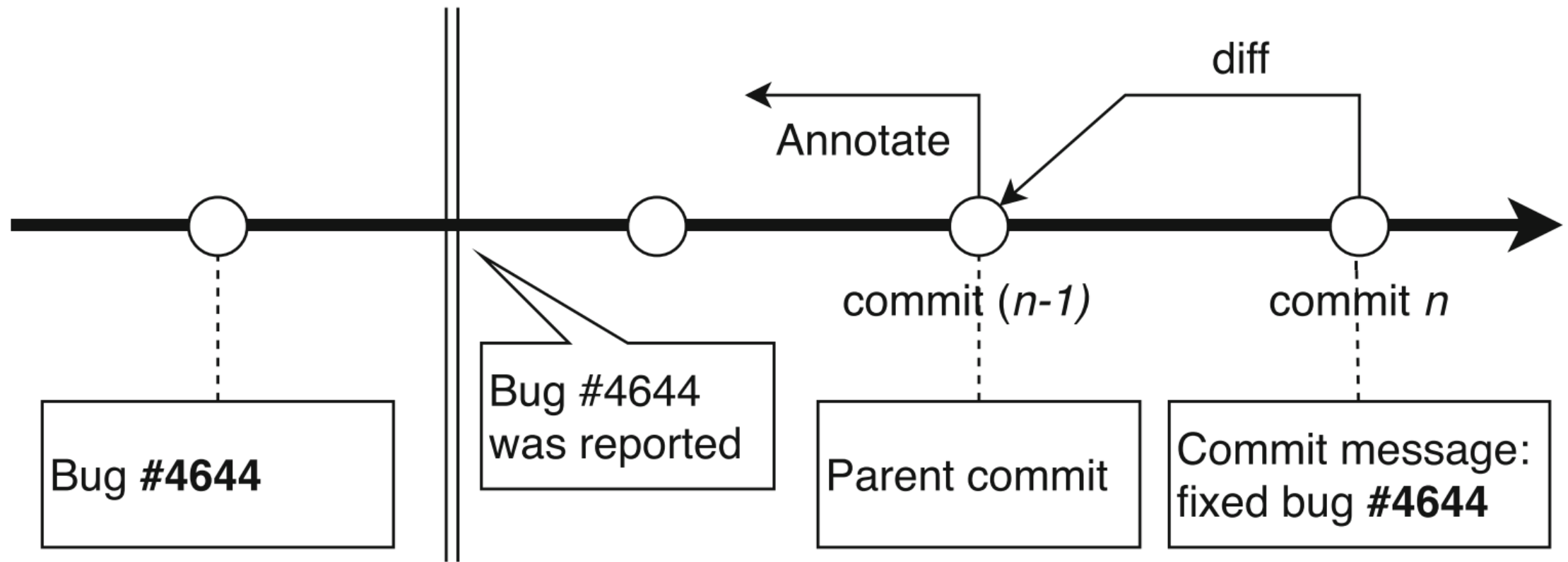


Top 3 purposes of using *diff*



Comparing *diff* outputs in 3 applications





SZZ: Locating bug-introducing changes

Recap of SZZ: Collection of Bug-fix Commits in **minecpp**

The collection of bug-fixing commits involved multiple strategies:

- Commit messages were analyzed to identify bug fix commits, leveraging keywords related to bug fixes. A total of **52 keywords** are used.
- Also, **Regular expressions** were utilized to perform precise searches for bug fix commit patterns
- The **SZZ algorithm** was applied to trace back and identify the buggy files associated with bug fix commits

“fixed ”, “ bug”, “fixes ”, “fix ”, “ fix”, “ fixed”, “ fixes”, “crash”, “solves”, “ resolves”, “resolves ”, “ issue”, “issue ”, “regression”, “fall back”, “assertion”, “coverity”, “reproducible”, “stack-wanted”, “steps-wanted”, “testcase”, “failur”, “fail”, “npe ”, “ npe”, “except”, “broken”, “differential testing”, “error”, “hang ”, “ hang”, “test fix”, “steps to reproduce”, “crash”, “assertion”, “failure”, “leak”, “stack trace”, “heap overflow”, “freez”, “problem ”, “ problem”, “ overflow”, “overflow ”, “avoid ”, “ avoid”, “workaround ”, “ workaround”, “break ”, “ break”, “ stop”, “stop ”

Commit messages with
52 keywords [6] (c.f. Sec. III-C)

```
+ RegEx. [7]: ‘.((solv(ed|es|e|ing))  
|(fix(s|es|ing|ed)?))’  
|((error|bug|issue)(s)?)).’
```

→ SZZ Algorithm [8]

[J. **S**liwerski, T. **Z**immermann, and A. **Z**eller; (2005)]

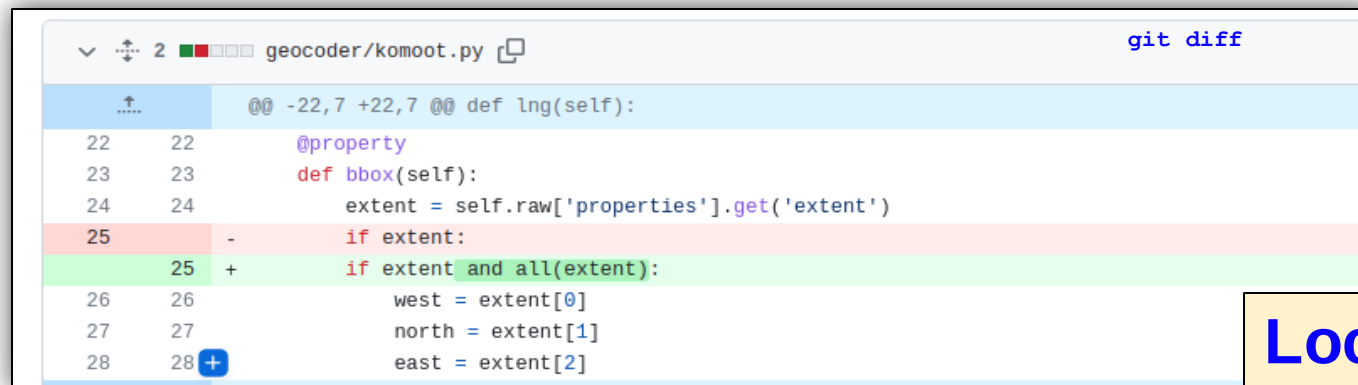
SZZ takes a commit (**bug-fixing**) as the input and returns a list of commits (**bug-introducing**) that last **added** the **deleted** lines in the input commit.

Recap of (fixed-1, fixed): Extracting Code Snippets in minecpp

The extraction of code snippets is done as follows:

Step 1:

- **Location:** The location of bug is found using **git diff** between **fixed - 1** and **fixed commit**. The location here indicates the **line number** in which the bug is detected and fixed.



```
git diff
@@ -22,7 +22,7 @@ def lng(self):
22 22         @property
23 23         def bbox(self):
24 24             extent = self.raw['properties'].get('extent')
25 -         if extent:
25 +         if extent and all(extent):
26 26             west = extent[0]
27 27             north = extent[1]
28 28             east = extent[2]
```

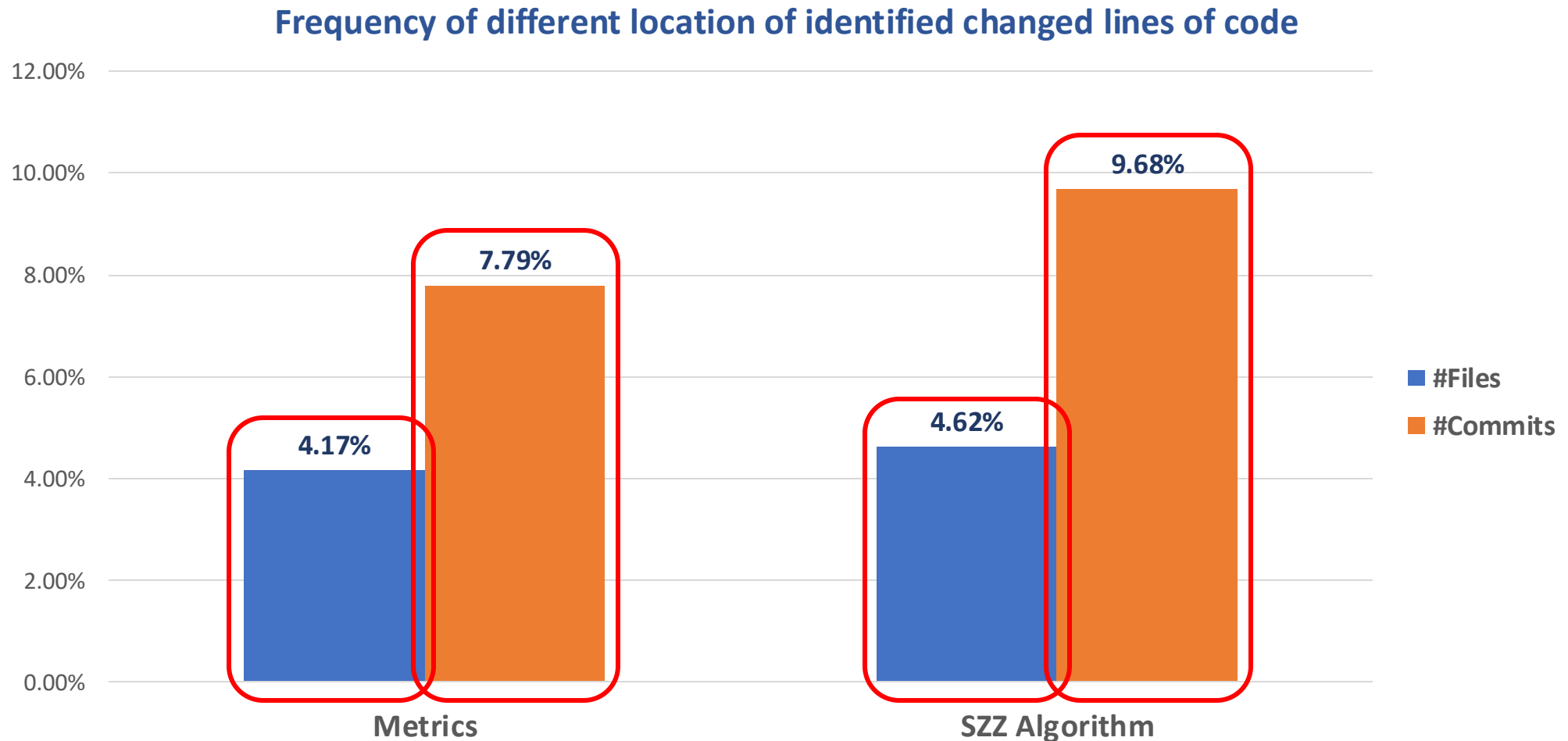
a **modification**
typically involves an
addition after a **deletion**

Loc: [before: 25, after: 25]

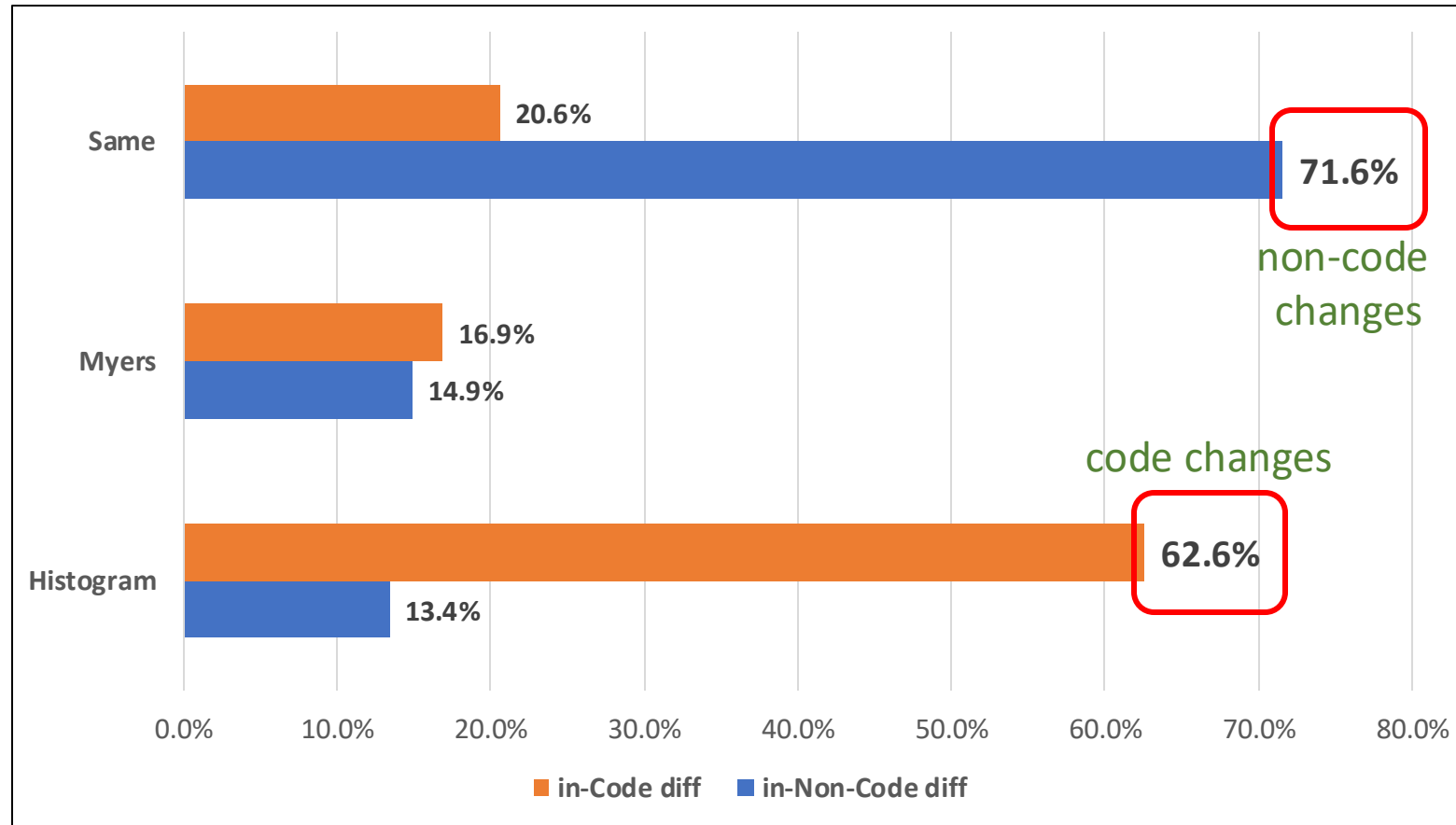
Bug Type: fix komoot.py when extent is none

Commit Message: fix calls to BBox with invalid input

Different *diff* algorithms can report different location of identified changed lines of code



Histogram is better for describing code changes



Use *histogram* diff algorithm when analyzing code changes

- Different diff algorithms can produce different amount and location of changed lines.
- Histogram detects the changed lines more appropriately from source code.



So, let's put Histogram to
git diff when mining code histories



Application on actual tools

- Git extension -- (Feature request)

<https://github.com/gitextensions/gitextensions/issues/6991>

The screenshot shows a GitHub issue page for 'Support diff algorithm "histogram" #6991' in the 'gitextensions/gitextensions' repository. The issue is marked as 'Closed' and was opened by 'gerhardol' on August 6, 2019. A comment from 'gerhardol' dated August 6, 2019, is visible. The comment includes a 'Feature description' section stating that Git supports four different algorithms in 'git-diff':

- myers (default)
- minimal (improved myers)
- patience (try to give contextual diff)
- histogram (kind of enhanced patience, normally faster too)

A link is provided: <https://git-scm.com/docs/git-diff#Documentation/git-diff.txt---diff-algorithmpatienceminimalhistogrammyers>. It also states 'All supported in at least since Git 1.7.7'.

A red box highlights a comparison link: 'A comparison of myers vs histogram here: https://www.researchgate.net/publication/330955835_How_Different_Are_Different_diff_Algorithms_in_Git_Use_--_histogram_for_Code_Changes'.

Below this, an example is given: 'Example myers vs patience. <https://gist.github.com/roryokane/6f9061d3a60c1ba41237>'. The comment concludes with: 'GE supports --patience argument (added just before histogram was added). Of course, a custom algorithm can be configured if patience is not ticked: \$ git config --global --add diff.algorithm histogram'.

- Pydriller

<https://pydriller.readthedocs.io/en/latest/configuration.html#git-diff-algorithms>

Minecpp depends on Pydriller

The screenshot shows the 'Git Diff Algorithms' section of the Pydriller documentation. It lists four algorithms: Myers (default), Minimal (improved Myers), Patience (try to give contextual diff), and Histogram (kind of enhanced patience). A link 'Differences between four diff algorithms' is provided.

A red box highlights a paragraph: 'Based on the comparison between Myers and Histogram in a study by Nugroho, et al (2019), various diff algorithms in the git diff command produced unequal diff outputs. From the result of patches analysis, they found that Histogram is better than Myers to show the changes of code that can be expected to recover the changing operations. Thus, in this tool, we implement histogram diff algorithm to consider differences in source code.'

Texts, References, and Acknowledgements

Online:

- Continuous Integration and Delivery (**CircleCI**: <https://circleci.com>)

Textbook:

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- Watson, K., Nagel, C., Pedersen, J. H., Reid, J. D., & Skinner, M. (2008). *Beginning Microsoft Visual C# 2008*. John Wiley & Sons.
- Mark J. Price (2024). *C# 13 and .NET 9 – Modern Cross-Platform Development Fundamentals*, 9th edition, Packt Publishing Ltd.

Reference:

- Soni, M. (2016). *DevOps for Web Development*. Packt Publishing Ltd.
- Yusuf Sulisty Nugroho, Hideaki Hata, and Kenichi Matsumoto. 2020. *How different are different diff algorithms in Git? Use --histogram for code changes*. Empirical Softw. Engg. 25, 1 (Jan 2020), 790–823.